

Akshay Shivashankar Bharadwaj

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Education

- University of Southern California**, MS in Computer Science (GPA 3.86) Jan 2025 – Dec 2026
- **Courses:** Analysis and Design of Algorithms, Web Technologies, Operating Systems, Cloud and Distributed Systems
- BMS College of Engineering**, BE in Computer Science (GPA 4.0) Aug 2018 – Jun 2022
- **Courses:** Data Structures and Algorithms, Web and Database design, Artificial Intelligence, Machine Learning

Experience

- IT Engineer**, Qualcomm - Bengaluru, India June 2022 – Dec 2024
- Engineered an **AngularJS-Flask** internal tool integrating **ServiceNow APIs** to automate end-to-end RFC (request for change) workflows, eliminating **100+ engineering hours of manual effort per month**.
 - Built end-to-end containerized microservices infrastructure using **Docker, Kubernetes (EKS)**, and AWS CloudFormation (S3, RDS, Route53, VPC, Lambda), cutting deployment time by **50%** and eliminating environment-parity issues.
 - Drove **cross-functional collaboration** with Dev, QA, and Infra teams to implement **CI/CD pipelines** via CloudFormation, replacing manual deployments and reducing environment setup errors to **near-zero**.
 - Designed an **AWS CloudFormation automation** framework to onboard **70+ web applications** with MFA and 24/7 CloudWatch/DataDog alerting to enforce security and availability SLAs and leveraged **GenAI tools** to drive cloud cost optimization by over **60%**.
- Interim Intern**, Qualcomm - Bengaluru, India Feb 2022 – Jun 2022
- Built a **self-service AWS deployment infrastructure using CloudFormation and Lambda** that enabled teams to deploy Java applications with minimal manual input, cutting new environment setup from days to **under an hour**.

Projects

- Operating Systems Kernel Development — Weenix (Unix-Like OS Kernel)** Aug 2025 – Dec 2025
- **Designed and implemented core kernel subsystems** in C, including process and thread management (scheduling, context switching, synchronization), virtual file system (VFS), and system calls within a Linux-inspired OS kernel.
 - **Developed virtual memory and process creation mechanisms**, implementing virtual memory mapping, page fault handling, and fork()-based child process creation to support concurrent user processes and robust memory isolation.
- Raft Consensus Protocol — Distributed Systems (C++ +20, gRPC)** Jan 2026 – May 2026
- Built a production-style implementation of the **Raft consensus leader election and log replication protocols** ensuring safety and liveness under node crashes, network partitions, and delayed RPC delivery, achieving **99.9% uptime** across simulated failure scenarios.
 - Designed concurrent, thread-safe state management for distributed RPC handlers and validated via fault-injection tests achieving **zero data loss** and **sub-500ms leader re-election** under real-world failure conditions.
- Artsy: Web & Mobile App using Django, PyTorch, Angular.js, Android Studio** Jan 2025 – May 2025
- Built a full-stack web and mobile app using AngularJS and Android Studio with a Django REST Framework API, PostgreSQL backend, and GCP deployment, achieving **sub-200ms API response times**.
 - Built a PyTorch image classification pipeline achieving **87% accuracy** across artwork styles; scraped and normalized artist metadata using BeautifulSoup to enrich API responses.
- Eye-Move: Eye-Tracking Based Typing Application (Computer Vision)** Oct 2021 – Jun 2022
- Leveraged Python and its Computer Vision libraries, mainly **OpenCV and DLib** for eye blink detection and creation of a virtual keyboard to help motor-neuron disability patients with communication.
 - Achieved **95% typing accuracy** through precise eye blink detection algorithms, enabling reliable and efficient communication.
- Moonquake Alert System (MAS), Space@yourService - EPFL (Swiss Federal Institute of Technology)** Jul 2020 – May 2021
- Built a **Python-based** data science application for moonquake detection leveraging seismic data from lunar rovers, achieving **92% detection accuracy** using signal processing and anomaly detection algorithms.
 - Coordinated with EPFL university to deploy on **Docker containers**, enabling real-time evacuation alerts with **sub-5 second response time** as part of the 'Asclepios' moon base establishment experiment SOPs.

Certifications

- AWS Certified Solutions Architect – Associate**, Amazon Web Services Issued: 2023
- Generative AI for Beginners**, Microsoft Issued: 2026

Technologies

Languages: Python, C++, C, Java, SQL, JavaScript, TypeScript, PHP, HTML, CSS, Unix Shell Scripting

Core Technologies: AWS, Git, Docker, Kubernetes, Tomcat, Flask, Django, PyTorch, REST APIs, AngularJS, Bootstrap, MongoDB, MySQL, Linux Kernel Programming, PostgreSQL, Bash scripting, Terraform, Jenkins, Ansible, Datadog, PowerBI, PagerDuty